

- Q.25 Describe the concept of reversibility Cycle.
- Q.26 What are the pre-starting checks of an engine?
- Q.27 What is the need of air cleaner? Also explain the different types of air cleaners.
- Q.28 Describe the heat balance sheet performance of an engine.
- Q.29 Explain the working of different components of fuel system in diesel engine.
- Q.30 Write a least five difference between gasoline engine and diesel engine.
- Q.31 What is Wankle Engine. Explain its working.
- Q.32 Discuss about lubrication system of IC engine.
- Q.33 What is the description and working of fuel injector pump?
- Q.34 Describe any five components of an IC engine, their function and material of construction.
- Q.35 What do you mean by preventive maintenance and overhauling?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe the various components of water cooling system of diesel engine. Also write down their function with importance.
- Q.37 Explain the various operations of a Diesel cycle in detail with suitable P-V diagram.
- Q.38 What are different types of tests for testing of an IC engine? Briefly explain any one of them.

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Time : 3 Hrs.

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SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 _____ is the difference between indicated and brake power of an engine.
a) Drawbar power b) Frictional power
c) Air flow d) PTO power
- Q.2 In a diesel engine the fuel is indicated by
a) Injected Fuel
b) Spark
c) Ignition
d) Heat resulting from compressing air that is supplied for combustion
- Q.3 As per IS specifications the endurance test for engine is conducted for
a) 100 hours b) 200 hours
c) 300 hours d) 500 hours
- Q.4 In a four stroke cycle engine, the cam shaft runs at
a) Half the speed of the crankshaft
b) One quarter the speed of the crankshaft
c) Double the speed of the crankshaft
d) Four the speed of the crankshaft.

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- Q.5 The volumetric efficiency of petrol engine is:
 a) 30-40% b) 40-60%
 c) 60-70% d) 75-80%
- Q.6 Which of the following is a transmission dynamometer:
 a) Torsion dynamometer
 b) Belt dynamometer
 c) Froude's hydraulic dynamometer
 d) Prony brake dynamometer
- Q.7 In I.C. diesel engines, the thermostat valve is opened at the temperature from
 a) 70 to 75°C b) 75 to 80°C
 c) 80 to 82°C d) 85 to 90°C
- Q.8 Carbon residue test is associated to evaluate the
 a) Oil quality b) Engine efficiency
 c) Engine speed d) Lubricant's quality
- Q.9 The function of piston ring, is to prevent
 a) Ignition delay
 b) Pre-ignition
 c) Leakage of air from the combustion chamber
 d) All above
- Q.10 The temperature, to which oil is heated before the oil vapour ignites, when a flame is passed over it, is called.
 a) Ignition point b) Pour point
 c) Burning scale d) Flash point

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SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Bore is length of engine cylinder. (True/False)
- Q.12 In Otto and diesel cycle engine, compression and expansion process are adiabatic. (True/False)
- Q.13 A petrol engine draws a mixture of petrol and air during suction stroke. (True/False)
- Q.14 Piston displacement is also known ____.
- Q.15 BS norms stands for ____.
- Q.16 The power available at the crank shaft is called ____.
- Q.17 Biogas contains ____, ____, and ____.
- Q.18 In diesel engine, liquid fuel injected alone under high pressure of ____.
- Q.19 The compression ratio of SI engine varies from ____ to ____.
- Q.20 Types of air cleaners.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write down the applications and importance of IC engine in agriculture sector.
- Q.22 Draw the P-V diagram of an Otto Cycle along with its processes.
- Q.23 Discuss about pollution control in IC engine.
- Q.24 Explain the different types of alternate fuels used in IC engine.

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